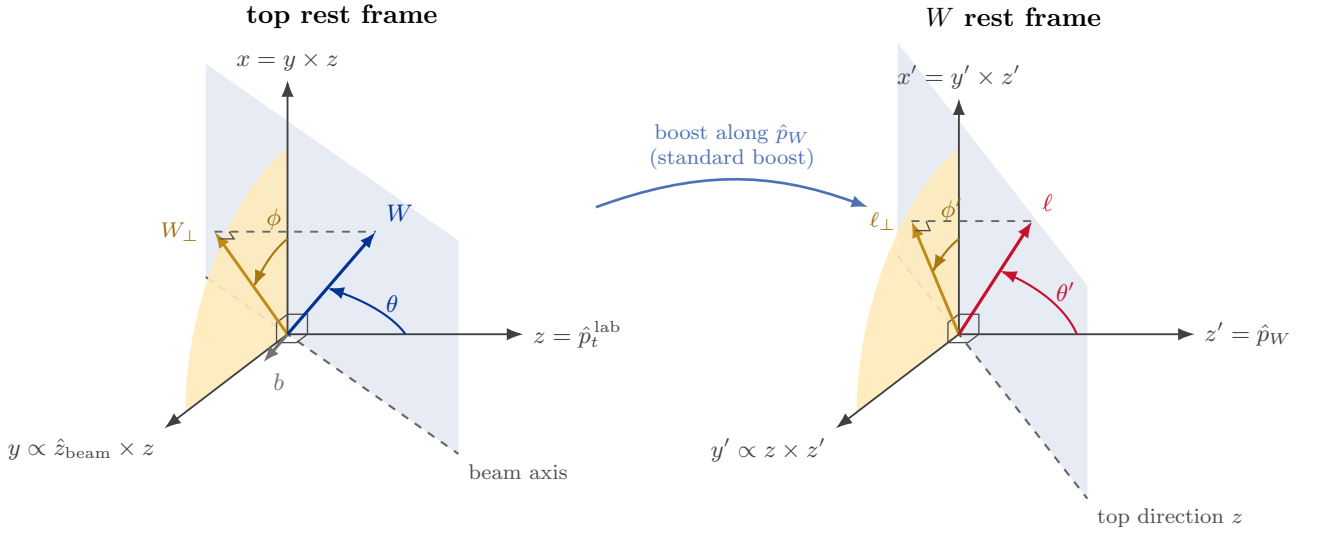


x–z plane (contains the beam and $\hat{p}_t^{\text{lab}}$; right: the x'–z' plane, contains $\hat{p}_W$ and the top direction) xy plane, perpendicular to z

θ: angle between z and the W. ϕ: **azimuth of the** W **around** z: project the W onto the xy plane ($W_\perp$); ϕ is the angle from x to $W_\perp$, toward y.
(θ', ϕ'): the same for the charged lepton in the W rest frame, with (x', y', z'): $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ (in the xy plane), $x' = y' \times z'$, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
The beam is not along z (z is the top direction); it lies in the x–z plane with a negative x component, since $y \propto \hat{z}_{\text{beam}} \times z$.
Hadronic top: same construction with the down-type quark in place of ℓ.